



GREY-BOX THERMAL MODELLING · MODEL STRUCTURE SELECTION

Systematic **xR+yC** structure selection for RC-based building digital twins

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From reduced-order models to a structure-selection gap

RC grey-box networks are established as DT thermal cores. What remains unresolved is how the topology itself is chosen.

ESTABLISHED

RC networks are the pragmatic core

Grey-box RC models sit between white-box physics simulation and black-box statistics: limited geometric input, preserved physical interpretability, low computational cost.

Li et al.; Boodi et al.; Serasinghe et al. [7, 8, 9]

KNOWN BUT UNRESOLVED

Topology drives accuracy and identifiability

Model structure controls prediction accuracy, computational cost and parameter identifiability. Oversimplified structures dominate error; complex ones increase runtime and correlation between parameters.

Chen et al.; Bacher & Madsen; Marty-Jourjon et al. [11, 12, 13]

THE GAP

No prescriptive xR+yC selection procedure

Reviews report **inconsistent naming and no standardised configurations**. In practice, the number of R and C — and how they are wired — remains an empirical choice.

Reported across [7, 8, 9, 11, 12]

THIS PAPER

A structured comparison workflow: build a bank of xR+yC candidates, calibrate them under one identical protocol on real operational data, and select on an **accuracy–parsimony criterion** — then state the limits of the resulting core.

The RC grey-box formulation

A lumped-parameter network: x thermal resistances, y thermal capacitances.

CONTINUOUS-TIME DYNAMICS – EQ. (3)

$$C \dot{T}(t) + K T(t) = G_{\text{out}} T_{\text{out}}(t) + G_q Q(t)$$

- $C = \text{diag}(C_1 \dots C_y)$ — thermal capacitances (stored energy). K is built from the resistances (heat transfer paths).
- Solved in state-space form and discretised exactly via the matrix exponential $A_d = e^{A\Delta t}$ — Eq. (6) — ensuring numerical stability for stiff thermal systems.
- Parameters $\theta = \{R_1 \dots R_x, C_1 \dots C_y, \alpha\}$ are estimated by minimising the quadratic prediction error on T_{in} — Eq. (7).

OPEN QUESTION

How many R ? How many C ? **Connected how?** Each answer is a different structural hypothesis about the building.

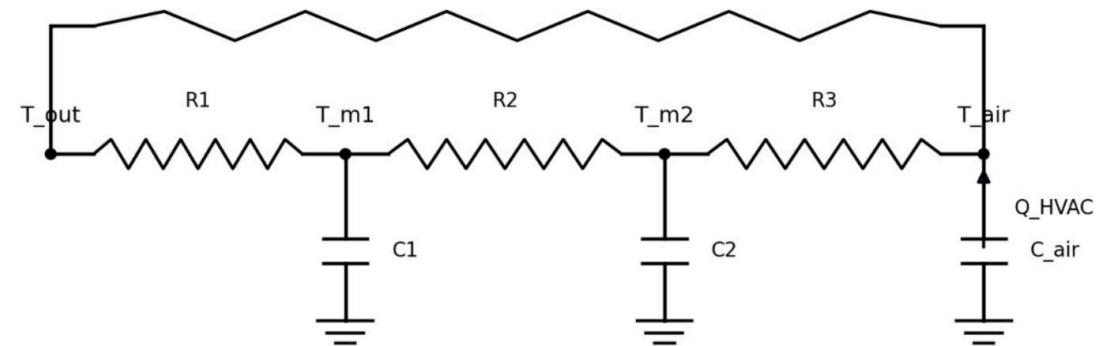


Fig. 2. Schematic of the grey-box RC thermal network (four resistors, three capacitors) — the structure ultimately selected in this study.

One year of real operational data — three measured signals

PLEIADData (Martínez Ibarra et al., 2023). Pleiades building, University of Murcia — Block A, hourly resolution.

CASE STUDY	DATA	MODEL BANK	SELECTION
PLEIADData — Pleiades building University of Murcia · Block A	1 year · hourly · 8 604 samples T_{in} · T_{out} · P_{HVAC}	19 candidate xR+yC structures 10 ladder · 9 ISO-inspired	Validation RMSE / MAE + AIC / BIC parsimony penalty



Fig. 1. Pleiades building — three blocks (A, B, C). This study is restricted to Block A; indoor temperature is the arithmetic mean over the available sensors.

- T_{in} indoor air temperature · T_{out} outdoor temperature · P_{HVAC} HVAC electrical consumption.
- Delivered thermal power is not measured.** It is mapped through an effective gain α , which aggregates COP variation, operating modes, distribution losses and auxiliary consumption.

EFFECTIVE THERMAL INPUT — EQ. (1)

$$Q(t) = \alpha \cdot P_{HVAC}(t)$$

α is **not** a fixed physical efficiency — an explicitly acknowledged simplification (Section 3.2).

TRAIN · JAN-SEP	VALIDATION · OCT-NOV	TEST · DEC
6 460 _h	1 464 _h	680 _h
Parameter estimation	Structure selection	Held out

PROTOCOL

Chronological split — no temporal leakage. Structures are ranked on data never seen during estimation.

A model bank of 19 candidates and one selection criterion

Two structural families are opposed; the identification protocol is held identical so that topology is the only variable.

FAMILY	STRUCTURES	STRUCTURAL HYPOTHESIS
LADDER	10 (1R1C → 10R10C)	Canonical series chains. Complexity added by stacking internal thermal mass.
STD (ISO-inspired)	9 (1R1C → 7R5C)	Physically structured: separate air node, distinct envelope masses, direct loss / shunt paths.

CONTROLLED Identical optimiser (L-BFGS-B), identical initialisation, identical log-space bounds, identical objective. **Only the topology varies.**

Fixed across all 19: $R = 0.2 \cdot C = 10^7 \cdot \alpha = 10^{-4}$; $\log R \in [-10, 5]$; $\log C \in [8, 25]$; $\log \alpha \in [-20, 2]$. Objective: MSE on T_{in} over the training set.

SELECTION CRITERION – EQ. (10)

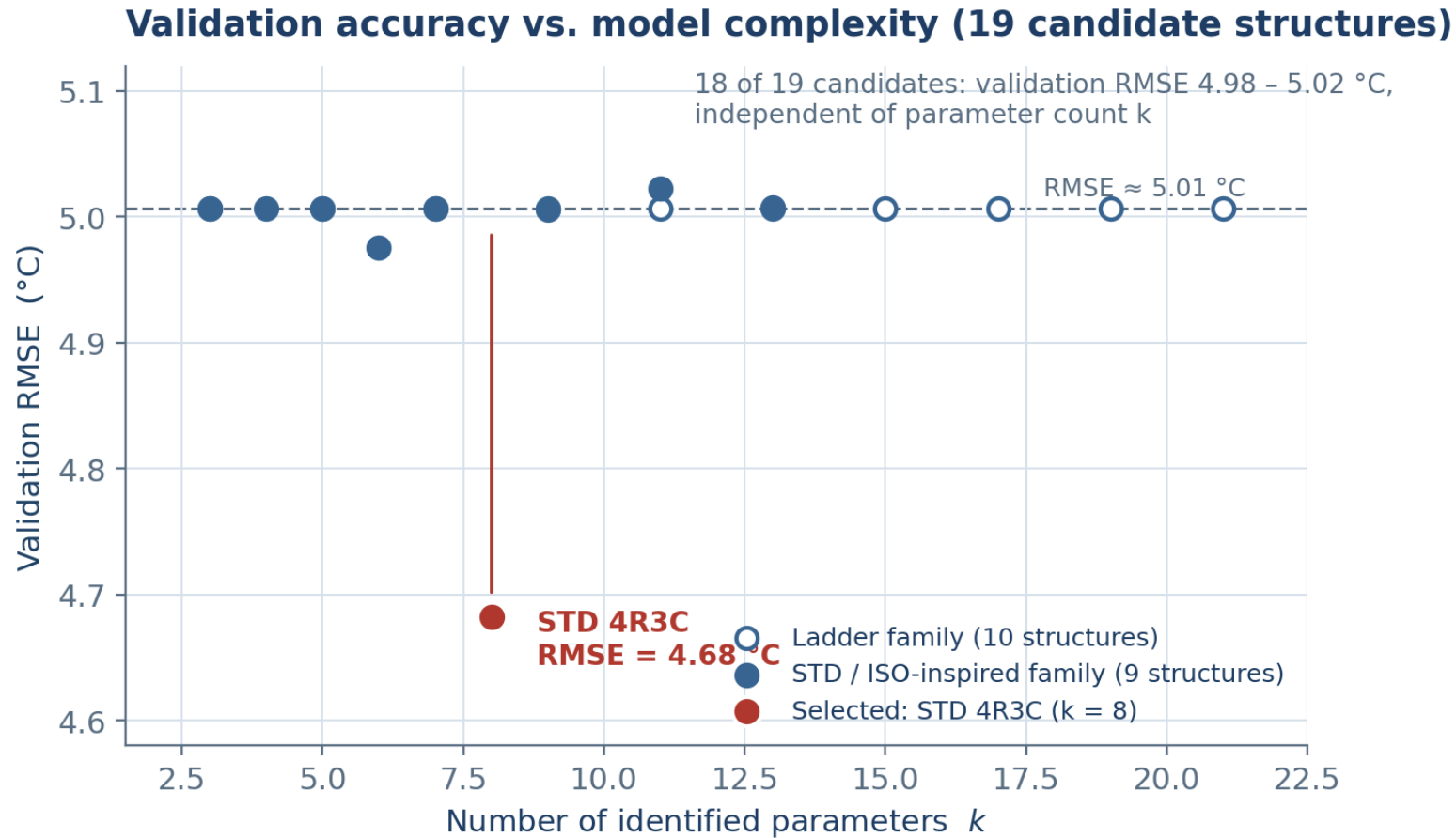
$$BIC = n \ln(RSS / n) + k \ln(n)$$

- **First term — fit.** Decreases as the residual sum of squares falls: a more accurate model lowers BIC.
- **Second term — penalty.** Grows with the number of estimated parameters k . Complexity is charged for.
- **Lower BIC = better accuracy–parsimony compromise.** A structure must improve the fit enough to pay for the parameters it adds.

CAVEAT AIC/BIC are used here as **penalised parsimony scores**, not as strict likelihood-based model evidence: residuals deviate from the i.i.d. Gaussian assumption under real operation (Section 3.8).

Additional RC states yield no systematic validation gain

Validation RMSE is essentially invariant across the model bank — except for one structure.



STD 4R3C – VALIDATION RMSE

4.68 °C

two masses + air shunt · $k = 8$

REMAINING 18 STRUCTURES

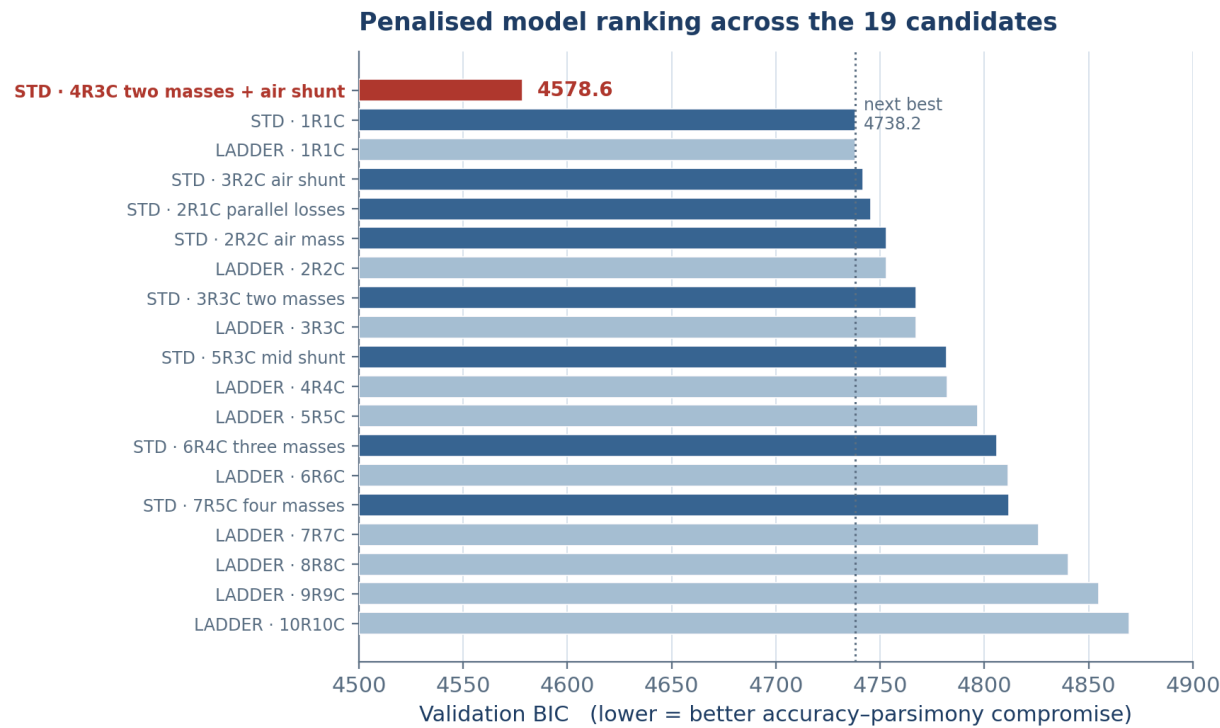
4.98 – 5.02 °C

Flat across $k = 3 \dots 21$

- The 10R10C ladder estimates **21 parameters** and does not outperform 1R1C with 3.
- Under the available excitation, deeper mass states are **not identifiable from the data** — they add parameters without adding information.

BIC ranks the bank and isolates one structure

Lower BIC = better compromise between fit and number of parameters.



STRUCTURE	K	RMSE	BIC
STD 4R3C	8	4.68	4578.6
LADDER 1R1C	3	5.01	4738.2
STD 3R2C air shunt	6	4.98	4741.7
LADDER 10R10C	21	5.01	4869.4

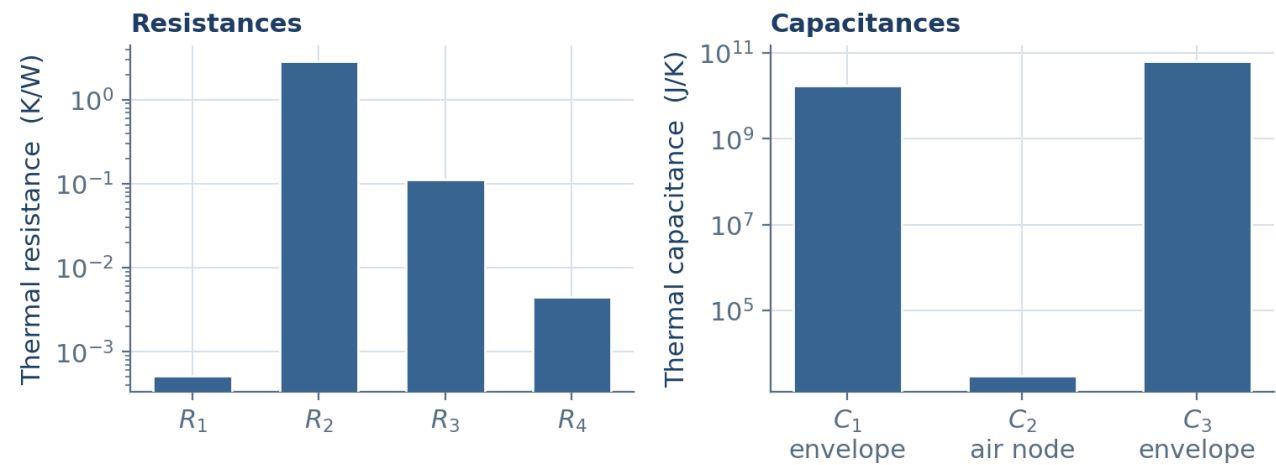
- The ladder family is penalised monotonically: each added state raises BIC without improving the fit.
- The selected structure is not the most complex — 8 parameters against 21 for 10R10C. It is selected on connectivity, not on order.

READING Its lower BIC combines **both** a lower RMSE and a moderate k — it is the only candidate that improves the fit enough to justify its parameters.

The selected structure separates the thermal time scales

Why this topology works, read from the identified values themselves.

Parameters identified for the selected STD 4R3C structure (log scale)



- Two large envelope capacitances ($C_1, C_3 \approx 10^{10}$ J/K) carry the **slow** thermal inertia of the building mass.
- One small air capacitance ($C_2 \approx 3 \cdot 10^3$ J/K) carries the **fast** indoor air response — seven orders of magnitude apart.
- The **air shunt** provides a direct low-resistance path between HVAC input and the air node, decoupling it from the slow envelope masses.

INTERPRETATION Ladder chains force all heat through a single series path. **4R3C is the only tested topology that separates fast and slow dynamics** — which is why it alone lowers the validation error.

Caution. Parameters are **effective lumped properties**, not component-level quantities: no geometric data was available to attribute them to specific envelope elements (Section 4.8).

Values identified from the three measured signals alone. Logarithmic scale — they span seven orders of magnitude.

A single global parameter set drifts across regimes

The selected structure is calibrated once, then run over the whole year.

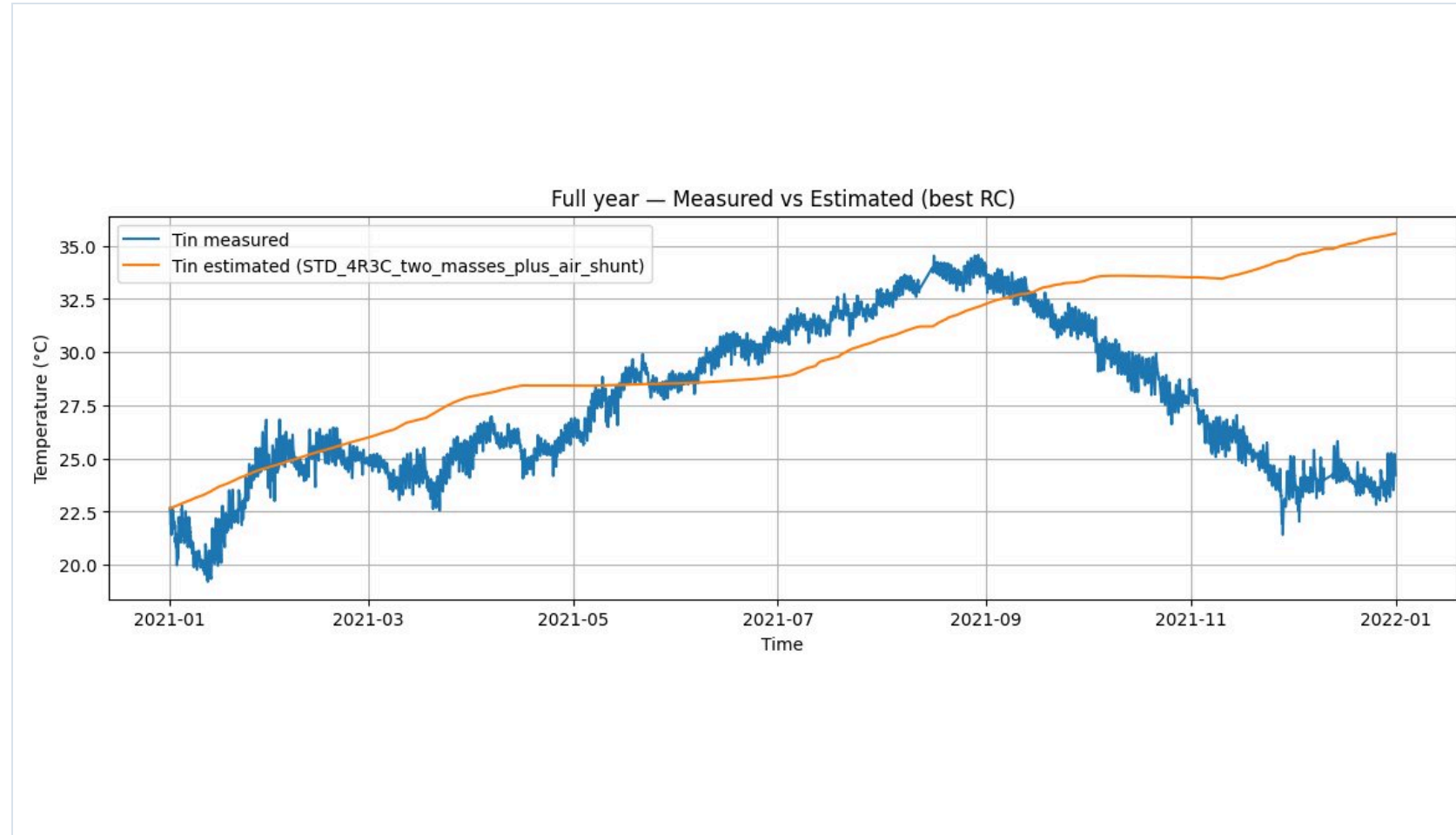


Fig. 3. Full-year measured vs. estimated indoor temperature — selected 4R3C model.

- **January → mid-summer:** the dominant seasonal trend is tracked.
- **From late summer:** a systematic positive bias appears progressively.
- The deviation is a **structural bias**, not high-frequency noise — it indicates modelling mismatch rather than stochastic error.

PLAUSIBLE CAUSE

Seasonal changes in HVAC effectiveness are **not represented by a constant α** ; a single global parameter set cannot span both regimes (Section 3.2).

The low December error is a scale effect, not superior generalisation

Test RMSE ≈ 0.86 °C. The paper argues explicitly against over-interpreting it.

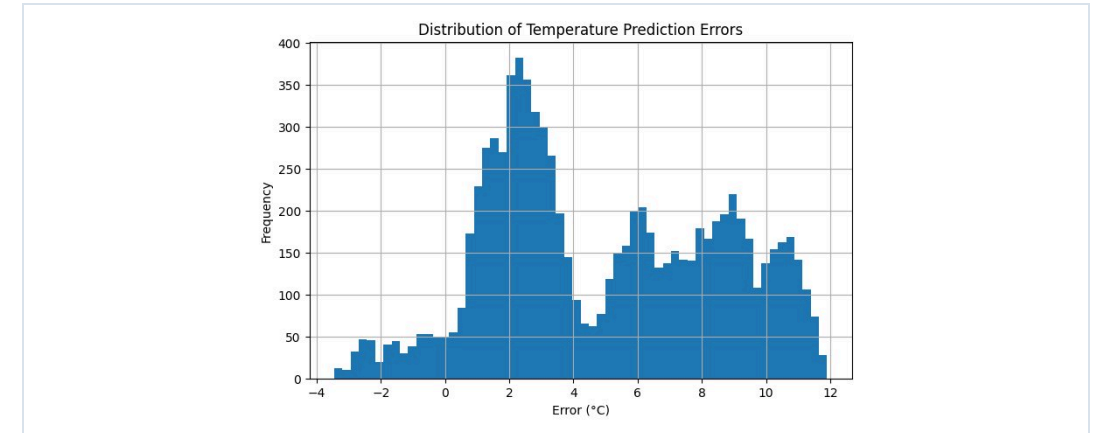
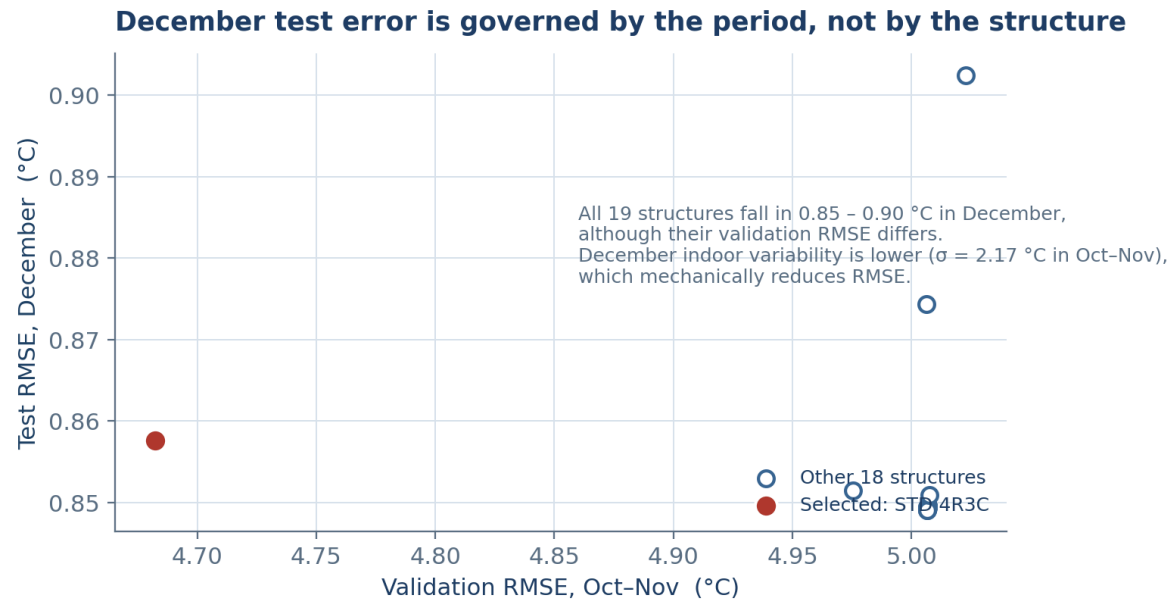


Fig. 5. Prediction errors over the full dataset — non-Gaussian, positively skewed, heavy positive tail.

WHY **RMSE is scale-dependent.** December indoor variability is lower than in the Oct–Nov transitional period; reduced variance mechanically reduces the absolute error, with the structural limitation unchanged.

THEREFORE Full-year behaviour and **residual structure** — not the December figure — are the primary evidence used to discuss the model's limits.

Conditions under which these results hold

Explicitly acknowledged in the paper (Sections 3.2, 4.6, 4.8).

- **No geometric information.** Without wall-by-wall areas, orientations and construction layers, R and C cannot be assigned to specific envelope components.
- **Parameters are effective lumped properties** — aggregated thermal behaviour, not physically separable wall elements.
- **Structural identifiability is limited.** Multiple parameter combinations may produce similar indoor responses under limited excitation.
- **α aggregates several mechanisms** and may partly compensate envelope parameters during calibration — a source of parameter ambiguity.
- **Single global parameter set.** Seasonal regime changes are not represented, producing the observed late-year bias.
- **Solar and internal gains are not explicitly modelled**, and are absorbed into the identified parameters.

SCOPE

These are the data conditions of the existing building stock — no BIM, no plans, no material data. The workflow is designed for, and evaluated under, precisely that constraint.

Structural complexity alone does not guarantee predictive performance

For the studied building and dataset, topology — not order — determines the quality of the RC thermal core.

FINDING 1

Order is not the lever

Beyond a moderate structure order, additional RC states yield **no systematic validation gain**, while increasing over-parameterization risk. Eighteen of nineteen candidates return a validation RMSE of 4.98 – 5.02 °C, from $k=3$ to $k=21$.

FINDING 2

Connectivity is the lever

A structured **4R3C topology** (two masses + air shunt) achieves the most favourable accuracy–parsimony compromise: RMSE 4.68 °C, BIC 4578.6, with only 8 parameters. Physically meaningful connectivity outperforms deep ladder states.

FINDING 3 & OUTLOOK

The boundary is seasonal, not structural

A single global parameter set drifts across regimes. Future work: **regime-dependent calibration**, integration of solar / internal-gain proxies, and geometric constraints to reduce lumping ambiguity — towards robust DT integration and MPC-oriented use.

CONTRIBUTION

A transparent, deployment-oriented workflow for selecting a parsimonious xR+yC topology under imperfect operational data — with its practical limitations stated explicitly.



Thank you for your attention

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